OC Core Methods and Reproducibility Companion v1.3.3

Methods, validation, reproducibility, and replay companion

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Dedicated to my dear wife Maria, without whom this work would have been impossible.

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AI Assistance Disclosure

AI-assisted tools were used as governed editorial and engineering instruments during preparation of this release package. Their roles were limited to bounded prose critique, formatting checks, source-path hygiene, figure and table QA support, and code/release-machine assistance under human review. They are not authors, do not provide scientific authority, and did not license any unsupported claim. The author retains full responsibility for the manuscript, evidence interpretation, claim boundaries, and final publication decisions.

Abstract

This methods companion explains how the formal, finite, numerical, and archival checks support the promoted claims. Many contemporary systems are too entangled to be understood by a single domain vocabulary: a technical platform carries physical infrastructure, software architecture, institutional incentives, security boundaries, user cognition, and economic feedback at the same time. The Ontology of Continua addresses that situation as a typed model-core project. It describes continuants, realizations, liveness, death, residue, boundaries, morphisms, operators, cycles, dimension, and K-level witnesses under explicit assumptions, so that a reader can ask how a system persists, changes, fails, and reappears without changing languages at every disciplinary border.

The project is intentionally ambitious in subject matter and intentionally bounded in claim discipline. Its publication task is not to replace physics, biology, cognition, systems engineering, enterprise architecture, or economics with a slogan. Its task is to offer a common structural grammar through which domain experts can compare state spaces, thresholds, flows, cycles, boundaries, and failure conditions without pretending that domain-specific science has become unnecessary.

Version 1.3.3 is a bounded external-review release. It is mature enough to be read as a coherent scientific manuscript and to be checked against formal and executable artifacts, but it remains a staged research program rather than a declaration of final all-domain completion. The release promotes model-core claims where their assumptions and evidence are visible; it leaves broader numerical closure, wider domain validation, and stronger comparative claims as future obligations.

The package contains a full monograph, a compact article route, a methods and reproducibility companion, a reviewer attack-and-response map, release navigation, public evidence anchors, finite-model outputs, formalization inventory, bounded retrospective replay QA, comparator and prior-art material, and appendices for proof, figures, tables, numeric rows, and source trace. The monograph is the canonical reading spine; the other documents are curated routes for first reading, editorial review, replay, and hostile criticism.

The evidence status is deliberately mixed and explicit. Some claims are supported by theorem routes, proof

sheets, Lean declarations, and finite semantic witnesses. Some claims are supported by bounded numerical or retrospective bounded replay rows that should be read as scoped QA evidence, not as complete empirical validation of a whole domain. Some claims are positioned against prior art or retained as research boundaries because the available evidence is not yet strong enough for a broader statement.

The principal limitation is therefore part of the manuscript's method: public wording must not outrun the evidence. OC Core 1.3.3 asks the reader to evaluate whether the typed model, proof governance, finite semantics, numerical anchors, figures, tables, appendices, and reviewer-response routes form a coherent bounded model-core contribution. Future work must add evidence or mechanization before it widens the public claim surface.

The practical value of the manuscript is therefore a form of disciplined compression. If the model is useful, it should help readers translate complex systems into a shared ontology while preserving the meaningful distinctions that make each domain scientifically serious. That is the standard by which the release asks to be read.

Version 1.3.3 Release Delta

The OC Core release sequence is part of a continuing scientific program rather than a series of isolated archive events. As the model is tested against formal criticism, domain examples, reproducibility checks, and external review, the public manuscript is expected to become more precise, more explicit about its limits, and more useful to readers outside the author's immediate working context.

A new release is therefore warranted only when there is a meaningful scientific or editorial delta: a clarified definition, stronger proof route, better evidence boundary, improved reader pathway, repaired notation, new domain projection, or more honest comparator position. Minor process movement is not enough. The public release should tell readers what changed in the scientific object and why that change matters.

This policy is also a commitment to regularity without pretending that research can be made mechanical. The author intends to make future public updates systematic enough for readers, reviewers, and auditors to follow the development of the model, while preserving the difference between a public scientific result and private working material. No unpublished working note or private editorial record is promoted as reader-facing evidence merely because it helped produce the release.

Version 1.3.3 is the release in which the OC Core corpus is reorganized from a set of scattered source witnesses and process records into a reviewable scientific package. The visible delta is not merely a new archive number: the release strengthens the typed foundation, makes the claim boundary explicit, integrates the theorem route with public proof sheets, and separates reader-facing prose from machine evidence.

The model delta includes clearer treatment of carriers, realizations, liveness, death, residue, rebirth, morphisms, generalized boundaries, hybrid operators, cycle modes, historical and effective dimension, and K-level witnesses. These additions are presented as a bounded model-core grammar rather than as an unrestricted theory of every phenomenon.

The verification delta adds and consolidates a Lean-oriented formalization inventory, structured proof sheets, finite semantic checks, finite witness interpretation, bounded retrospective replay QA, comparator and prior-art positioning, negative-control and falsifier language, and adversarial-review response material. Where a stronger claim is not yet earned, the release records the limitation instead of hiding it inside a source-control record.

The editorial delta is equally important. Figures, tables, formulas, numeric anchors, appendices, and evidence routes are kept in the publication corpus; machine coverage maps remain coverage and trace data only. The public documents are therefore built from the human manuscript hierarchy and curated payload sources, while source bindings and machine evidence indexes stay in the review manifest.

For a returning reader, the practical point is this: 1.3.3 should be read as a clarification and consolidation release. It does not claim final closure. It improves the public surface through which the model can be criticized, taught, checked, and extended.

Reader Routes

Dear readers, systems theory is of interest to many communities, but rarely in exactly the same way. A formal reviewer, a philosopher of systems, an applied architect, an AI engineer, a security specialist, and an executive reader may all ask legitimate questions of the same manuscript. OC Core 1.3.3 is therefore designed as a deliberately generous scientific reading surface: it aims to disclose the Ontology of Continua as an applied model of system architecture while giving different readers a courteous route into the material.

Scientific reviewers and formal critics may wish to start with the abstract, release delta, model-foundation route, theorem/proof integration route, formalization inventory, finite semantic witnesses, proof machinery, and falsifiability sections. This route is meant to make the work attackable in the best sense: every promoted claim should expose its assumptions, proof or executable evidence, counterexample boundary, and reopening condition.

Systems theorists, philosophers, and interested theoretical readers may prefer to start with the continuum problem, the historical and systems-theory motivation, the K-level primer, lifecycle and boundary chapters, prior-art comparison, and limitations. This path asks what OC inherits, what it reorganizes, where its residual delta begins, and where predecessors or parallel branches already carry part of the work.

Applied architects of complex systems–including engineers, enterprise architects, AI builders, security specialists, and applied researchers–may find it useful to start from practical examples, domain projections, figures, tables, and worked routes before returning to the formal core. The model chapters can then be read as a design grammar for carriers, realizations, boundaries, operators, update/flow separation, evidence classes, and claim boundaries.

CIOs, CEOs, enterprise architects, and strategy readers may sensibly read the release delta, practical utility chapter, model-comparison appendix, and final conclusion before investing in proof details. This route asks what the model is for, what it can support today, what remains too immature for operational commitment, and how evidence classes affect research or enterprise decisions.

Reproducibility auditors, evidence reviewers, and benchmark readers may go directly to the methods and evidence chapters, retrospective bounded replay QA, numeric tables, machine-readable table reference, audit trail, source-trace appendix, and public evidence package manifest. This route asks whether formula, input snapshot, comparator, residual or uncertainty, negative control, falsifier, replay hash, and failure interpretation are present.

The companion is useful when commands, inputs, outputs, checksums, finite witnesses, retrospective bounded replay, and failure interpretation are the main concern. In the full package, the conceptual model and formulas live in the monograph's scientific body; the proof route and formalization inventory live in theorem, proof, and formalization sections; numeric evidence and retrospective bounded replay QA live in the methods/evidence route; figures and tables live inline in the monograph; appendices carry source trace, proof machinery, comparison rows, audit trail, and machine-readable table references; prior-art comparison and reviewer objections live in the article and attack-response map.

The author asks all readers to treat limitations as part of the claim, not as a footnote. A statement is promoted only where its assumptions, proof or evidence class, comparator boundary, and reopening condition are visible. Claims about complete scientific coverage, unrestricted all-domain numerical closure, or unrestricted superiority over contemporary science are not promoted by this release.

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1 Methods and Reproducibility Companion

The methods companion explains how a reader can inspect OC Core 1.3.3 without treating machine-readable material as the manuscript itself. Reproducibility is used here as scientific interpretation: a check matters because it bears on a named claim, assumption, formula, witness, comparator, or falsifier.

1.1 What a Replay Checks

A replay row connects an input snapshot, a formula or reconstruction rule, an expected output, a residual or uncertainty, a comparator, a negative control, and a failure interpretation. Passing such a row supports the bounded claim named by the row. It does not by itself validate an entire domain or remove the need for broader empirical work.

1.2 Formal and Finite Checks

The proof route and finite semantic checks give the model a stricter ceiling. A theorem statement is supported only when the assumptions, proof sheet, dependency relation, mechanized subset where present, and counterexample boundary remain inspectable. A finite witness is valuable because it can show that a definition or operator behaves as claimed inside a bounded semantics.

1.3 Interpreting Failure

A failed check is not a formatting problem. It may reopen a claim, narrow a domain route, reveal a missing assumption, or show that a comparator already explains the case. The companion therefore treats mismatch, missing input, missing hash, drifted output, absent negative control, and unsupported extrapolation as scientific events.

1.4 Audit Trail and Machine-Readable Tables

The reader-facing tables summarize the evidence in prose. Machine-readable tables remain available for auditors who need to inspect source rows directly: claim rows, theorem rows, finite checks, retrospective bounded replay QA, numeric tables, comparator material, and source-trace material. The tables are support for the argument, not a substitute for it.

1.5 Retrospective Replay Path Integrity

The retrospective bounded replay table is a reader-facing reproducibility anchor, but it is not presented as an independently auditable prospective-blind study in this package. Its public path is printed exactly so that an auditor can compare the prose, the PDF extraction, and the machine-readable evidence package without guessing which file is meant:

```
validation/target_blind/OC133_TARGET_BLIND_PREDICTION_TABLE.json
```

This table is not a broad empirical triumph claim and not a prospective-blinding proof. It records bounded reconstruction rows, source references where available, reconstruction rules, comparators, negative-control conditions, and reopening conditions. Prospective blinded-study promotion requires checksum-bound split locks, raw snapshots, pre-run projection records, timestamps, and runner evidence in a future evidence package.

The directory name contains the historical phrase `target_blind`; the public claim does not rely on that path name. Until checksum-bound split locks, raw snapshots, pre-run projection records, timestamps, and runner evidence are present in the public package, this file is interpreted as retrospective bounded replay QA rather than as an independently auditable prospective-blind study.

1.6 Reader-Facing Audit Trail

The audit trail is read from the scientific claim outward. First, identify the claim and its declared assumptions. Second, inspect the formula or finite witness used to support it. Third, compare the output with the named source snapshot and comparator. Fourth, check the residual, uncertainty, negative control, and falsifier. Fifth, decide whether the public wording is still narrow enough. The companion keeps this order visible because a reproducible artifact is useful only when a reviewer can say what claim it supports and what failure would mean.

1.7 Public Reproducibility Protocol

The public replay route is intentionally procedural. A reviewer should identify the claim class, inspect the source snapshot, run or reproduce the bounded check, compare the expected output, and then decide whether the public wording remains justified.

Step	Reader action	Required public evidence
1	Locate the claim row	claim id, title, assumption set, promoted wording
2	Locate the input	source snapshot, hash, date or fixture identity
3	Locate the method	formula, reconstruction rule, finite witness, or proof sheet
4	Run or inspect the check	command, expected output, residual or theorem status
5	Interpret failure	negative control, comparator, falsifier, reopening condition

Representative command surfaces are intentionally printed as public commands, not private build instructions:

```
python tools/audit_oc_core_release_assembly_machine.py
  --release oc_core_1_3_3
  --assembly-revision oc_core_1_3_3_review_current
  --check

python tools/audit_oc_core_release_quality.py
  --release oc_core_1_3_3
  --assembly-revision oc_core_1_3_3_review_current
  --check

python -m pytest release_machine/tests/test_release_assembly_machine.py -q
```

The alias `oc_core_1_3_3_review_current` is the public review alias for the assembly identifier recorded in the review manifest. Internal recovery labels remain outside title pages and reader-facing identity surfaces.

The assembly metadata field `public_review_revision_aliases` binds the alias to the exact internal assembly revision, review ZIP, manifest hash, and artifact hashes used by the audit; reviewers should inspect that JSON binding rather than relying on prose.

1.8 Public Review Alias Binding

The alias binding is part of the package assembly metadata, not a title-page identity claim. A reviewer should check the following public fields:

Field	Expected role
`public_review_revision_aliases.oc_core_1_3_3`	resolves the public alias to the exact internal assembly revision and review package
`package/OC_CORE_RELEASE_PACKAGE_MANIFEST_1_3_3.json`	lists every packaged file with checksum
`package_assembly/OC_CORE_RELEASE_PACKAGE_ASSEMBLY_1_3_3.json`	records artifact paths, hashes, and audit state
scoped editorial Cerberus summary path named in the alias binding	records the scoped editorial review result for the same artifact hashes

The retrospective bounded replay file is the principal prediction-row anchor:

`validation/target_blind/OC133_TARGET_BLIND_PREDICTION_TABLE.json`

The replay result should be read as bounded support. It becomes stronger only when the table row carries source snapshot, formula, observed value or theorem result, comparator, residual or uncertainty, negative control, falsifier, and a clear failure interpretation.

One mathematics replay row is explicitly demoted in this release because the finite-model source currently records certificate/source-binding failures. That row is a demotion boundary, not a support row, until `source_manifest_binding_ok` is true and `certificate_binding_failure_total` is zero in the finite-model checks. This is intentional: the table must not promote a replay row that the underlying finite-check source cannot presently support.

1.9 Failure Interpretation Matrix

Failure class	Scientific meaning	Public response
Missing source snapshot	Evidence cannot be inspected	demote or suspend the claim
Hash mismatch	The row no longer identifies the same artifact	rerun and disclose drift
Formula mismatch	The method no longer supports the wording	narrow the claim
Comparator succeeds	OC may not add residual delta	update prior-art boundary
Negative control succeeds	The check is not discriminating	reopen the evidence route
Falsifier triggers	The promoted wording is false or too broad	retract, repair, or demote

2 Scientific Evidence Command And Output Matrix

The public reproducibility route is read from source claim to evidence lane, not from PDF production alone. The table below records the evidence commands, expected outputs, and failure meanings that bound the scientific claims in this package.

Evidence lane	Command or inspection route	Required input	Expected output	Failure meaning
Finite semantic checks	python proofs/finite_model_checks/run_finite_model_checks.py	proofs/finite_model_checks/OC133_FINITE_MODEL_INPUTS.json	[illegible] and proofs/finite_model_checks/FINITE_MODEL_REPL	[illegible] demotes the exact theorem or finite-witness sentence that depends on it.
Retrospective replay table	inspect validation/target_blind/OC133_TARGET_BLIND_PREDICTION_TABLE.json	locked source snapshot, formula, comparator, residual, negative control, and falsifier fields	bounded replay row with explicit support class	Missing source/formula/comparator/falsifier fields prevent empirical promotion.
Numeric replay QA	inspect validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json	numeric replay rows and QA status fields	row-level bounded replay verdict	A residual breach or missing negative control demotes the numeric claim.
Proof sheets	inspect proofs/proof_sheets/T133-*.md, proofs/THEOREM_REGISTRY_1_3_3.json, and proofs/PROOF_DEPENDENCY_GRAPH_1_3_3.json	theorem statement, assumptions, proof route, finite witness, reopening condition	checksum-bound proof-route record	A missing proof sheet or dependency mismatch demotes theorem wording.
Formalization inventory	inspect formal/lean/OC133V12 and formal/lean/LEAN_BUILD_CERTIFICATE_1_3_3.json	Lean source, certificate state, source manifest binding	formalization inventory with explicit certificate boundary	If certificate binding is not clean, Lean remains source inspection rather than promoted machine-checked support.
Prior-art and comparator boundary	inspect docs/OC_1_3_3_PRIOR_ART_COMPARATOR_MATRIX.json and comparators/OC_1_3_3_COMPARATOR_MATRIX.md	comparator rows, source snapshots, novelty boundary	bounded residual-delta statement	A stronger comparator narrows or removes the OC novelty sentence.

These rows are deliberately conservative. They tell a reviewer where a claim can be checked and what kind of failure would change the public wording.

Keywords and Citation Route

Keywords: Ontology of Continua; continuum ontology; typed model core; systems theory; autopoiesis; dynamical systems; hybrid systems; formal methods; formalization inventory; finite semantic checks; retrospective bounded replay QA; reproducible research; artifact evaluation; claim governance; falsifiability; evidence-bound scientific publishing.

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Open repository: https://github.com/alexanderyashin/ontology-of-continua-core-main. Repository files support reproducibility and source inspection; they do not replace the manuscript's claim boundaries.